

Waddah Daker

+1 (514) 885-5443 waddah.daker@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

Education

Concordia University

Bachelor of Computer Science (GPA: 3.6/4.3)

Expected Jan 2027

Montréal, QC

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Databases

American University of Sharjah

Bachelor of Computer Science (GPA: 3.9/4.0)

Sep 2022 – Apr 2023

Sharjah, UAE

Work Experience

Software Developer

AW Design (AWD Decoration Design & Fit Out L.L.C.)

May 2026 – Present

Remote

- Shipped a bilingual Next.js/TypeScript site with 15+ Sanity CMS schemas and RTL/LTR i18n
- Designed a serverless AWS recruitment portal (Lambda, DynamoDB, SES) for 3 branches, targeting <\$5/mo
- Set up the firm's AWS account from scratch with MFA-secured root and role-based, least-privilege IAM access

Projects

Harmful Algal Bloom RL Simulation (AlgaeMind) | [GitHub](#) | [DevPost](#) | *React, TypeScript, FastAPI, PyTorch*

- Developed heuristic, Q-learning and Claude-based mitigation agents; across 20 randomized 60-day scenarios, the heuristic agent cut bloom coverage by 99.8% and raised lake health from 56 to 91
- Modeled lake dynamics with a Python/FastAPI cellular-automata simulation on a 560-cell grid at 174 ticks/s, covering 60 days in under 1.5 s
- Served grid snapshots through a REST API at 3.4 ms p50 / 9.6 ms p95 (337 req/s), rendered live in React on HTML5 Canvas at a steady 60 FPS

Reliable Data Transfer Protocol over UDP | [GitHub](#) | *Java, UDP Sockets, Linux*

- Built a reliable data-transfer protocol from scratch in Java over raw UDP (packet serialization, alternating-bit sequencing, timeout-driven retransmission), achieving 100% file transfer success across all 4 test scenarios
- Stress-tested the protocol under 4 adversarial network conditions (100ms/1100ms delay, 50% client/server packet loss) using Linux network namespaces and tc netem fault injection

Reuters SPIMI Indexer and Search Engine | [GitHub](#) | *Python, BeautifulSoup, Information Retrieval*

- Indexed 21.5K Reuters documents (2.8M tokens) into a 1.6M-posting inverted index for a Boolean search engine, answering AND/OR queries in under 0.1 ms
- Implemented a SPIMI indexer that builds the index $6.8\times$ faster than a sort-based one (12.5 s $\rightarrow$ 1.85 s)
- Cut the dictionary by 37.5% and postings by 30% by removing punctuation, numbers and stopwords

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, FastAPI, Pandas, NumPy, PyTorch, Scikit-Learn

Tools & Technologies: Git, Docker, AWS, Vercel, Sanity CMS, Linux, PostgreSQL, MySQL, Vite, Postman, Wireshark

Certifications: AWS Certified Solutions Architect – Associate (2026)

Spoken Languages: Arabic (Native), English (Fluent), French (A1)